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QUESTION 1 COMMENTING

```
% DO NOT REMOVE THE LINE BELOW
% MAKE SURE 'eel3135_lab06_comment.m' IS IN THE SAME DIRECTORY AS THIS FILE
clear; close all; clc;
type('eel3135_lab08_comment.m')

%% USER-DEFINED VARIABLES
clear
close all
clc

%% DEFINE FILTER
N = 10;
h = (1/N)*ones(N,1);

% What is the impulse response of this filter?
% Use d in place of delta.
% 1/10 (d[n]+d[n-1]+d[n-2]+d[n-3]+d[n-4]+d[n-5]+d[n-6]+d[n-7]+d[n-8]+d[n-9])
LOWPASS

% COMPUTE THE DTFT
n = 0:(N-1);
w = -pi:pi/5000:pi;
H = DTFT(h,w);
```

```

% PLOT THE IMPULSE RESPONSE AND DTFT
figure
subplot(3,1,1)
stem(n,h)
xlim([-0.5 20.5])
title('Impulse Response of h')
xlabel('Time Index (n)')
ylabel('Amplitude')
subplot(3,1,2)
plot(w,abs(H))
grid on;
title('Magnitude Response of H')
ylabel('Magnitude [rad]')
xlabel('Normalized Angular Frequency [rad/s]')
subplot(3,1,3)
plot(w,angle(H))
grid on;
title('Phase Response of H')
ylabel('Phase [rad]')
xlabel('Normalized Angular Frequency [rad/s]')

function H = DTFT(x,w)
% ==> Computes the DTFT of the signal at frequencies w <==

    H = zeros(length(w),1);
    for nn = 1:length(x)
        H = H + x(nn).*exp(-1j*w.*(nn-1));
    end

end

```

QUESTION 2: DTFT OF COMMON FUNCTIONS

```

T = pi/10;

n = 0:19;
w = linspace(-pi, pi, 500);

```

2 (a) PLOT DTFT

ALSO ANSWER: Is the data predominantly low frequency, high frequency, or neither?

(a) $x[n] = \delta[n]$

```

x_a = [1, zeros(1,19)]; %  $\delta[n]$ , impulse at  $n = 0$ 
H_a = DTFT(x_a, w);

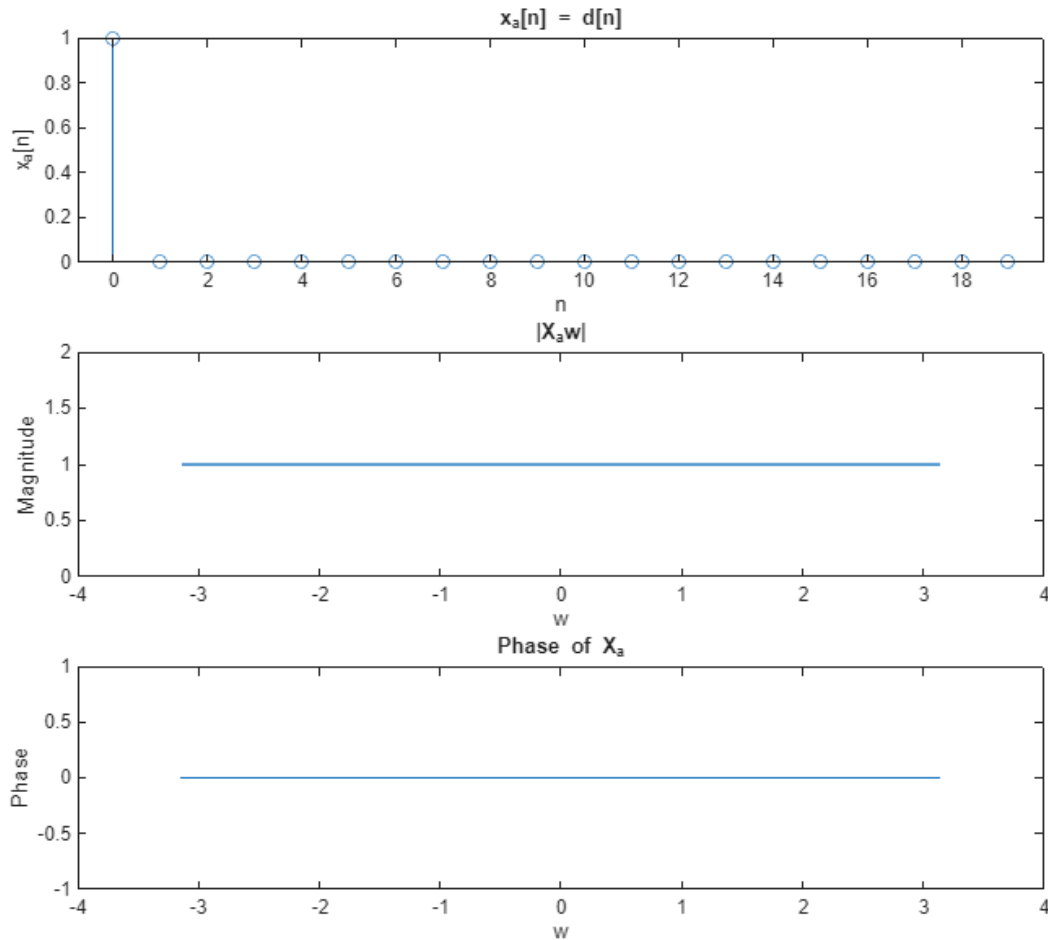
figure;
subplot(3,1,1); stem(n, x_a); title('x_a[n] = d[n]'); xlabel('n');
ylabel('x_a[n]');

```

```

subplot(3,1,2); plot(w, abs(H_a)); title('|Xaw|'); xlabel('w');
ylabel('Magnitude');
subplot(3,1,3); plot(w, angle(H_a)); title('Phase of Xa'); xlabel('w');
ylabel('Phase');

disp('(a) delta[n] is neither high or low frequency.');
```



2 (b) PLOT DTFT

ALSO ANSWER: Is the data predominantly low frequency, high frequency, or neither?

```

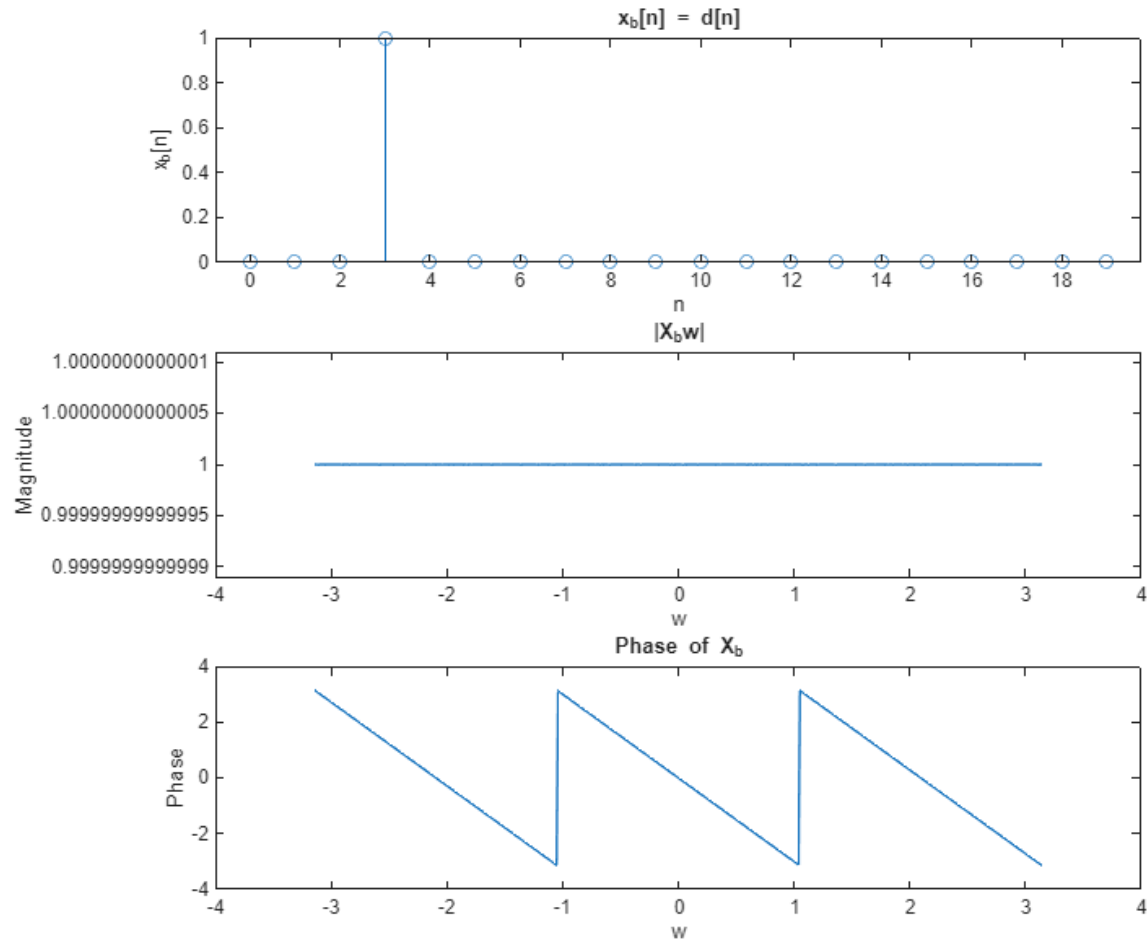
x_b = [zeros(1,3), 1, zeros(1,16)]; %  $\delta[n-3]$ 
H_b = DTFT(x_b, w);

figure;
subplot(3,1,1); stem(n, x_b); title('xb[n] = d[n]'); xlabel('n');
ylabel('xb[n]');
subplot(3,1,2); plot(w, abs(H_b)); title('|Xbw|'); xlabel('w');
ylabel('Magnitude');
subplot(3,1,3); plot(w, angle(H_b)); title('Phase of Xb'); xlabel('w');
```

```

ylabel('Phase');
disp('(b) delta[n-3] is neither low nor high frequency.');
```

(b) $\delta[n-3]$ is neither low nor high frequency.



2 (c) PLOT DTFT

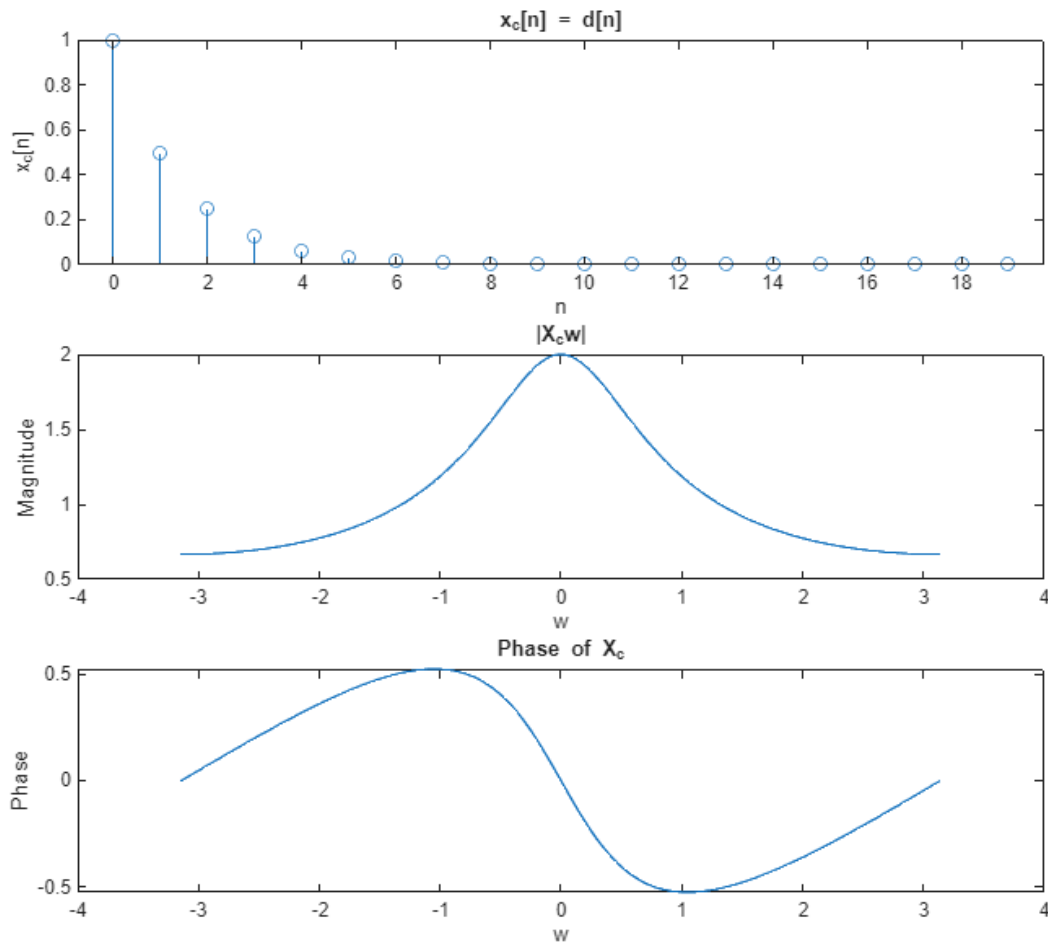
ALSO ANSWER: Is the data predominantly low frequency, high frequency, or neither?

```

x_c = (1/2).^n;
H_c = DTFT(x_c, w);

figure;
subplot(3,1,1); stem(n, x_c); title('x_c[n] = d[n]'); xlabel('n');
ylabel('x_c[n]');
subplot(3,1,2); plot(w, abs(H_c)); title('|X_c[w]|'); xlabel('w');
ylabel('Magnitude');
subplot(3,1,3); plot(w, angle(H_c)); title('Phase of X_c'); xlabel('w');
ylabel('Phase');
disp('(c) This signal is predominantly low frequency.');
```

(c) This signal is predominantly low frequency.



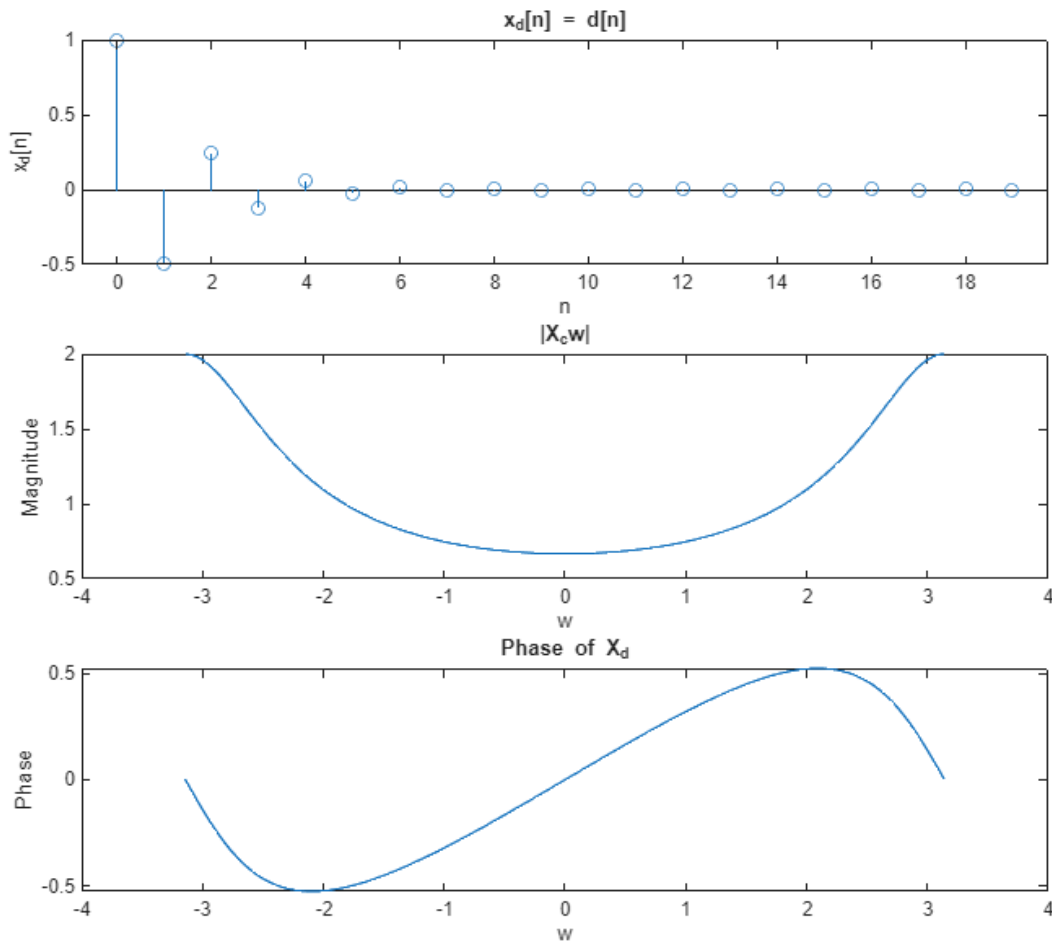
2 (d) PLOT DTFT

ALSO ANSWER: Is the data predominantly low frequency, high frequency, or neither?

```
x_d = (-1/2).^n;
H_d = DTFT(x_d, w);

figure;
subplot(3,1,1); stem(n, x_d); title('x_d[n] = d[n]'); xlabel('n');
ylabel('x_d[n]');
subplot(3,1,2); plot(w, abs(H_d)); title('|X_cw|'); xlabel('w');
ylabel('Magnitude');
subplot(3,1,3); plot(w, angle(H_d)); title('Phase of X_d'); xlabel('w');
ylabel('Phase');
disp('(d) This signal is predominantly high frequency.');
```

(d) This signal is predominantly high frequency.



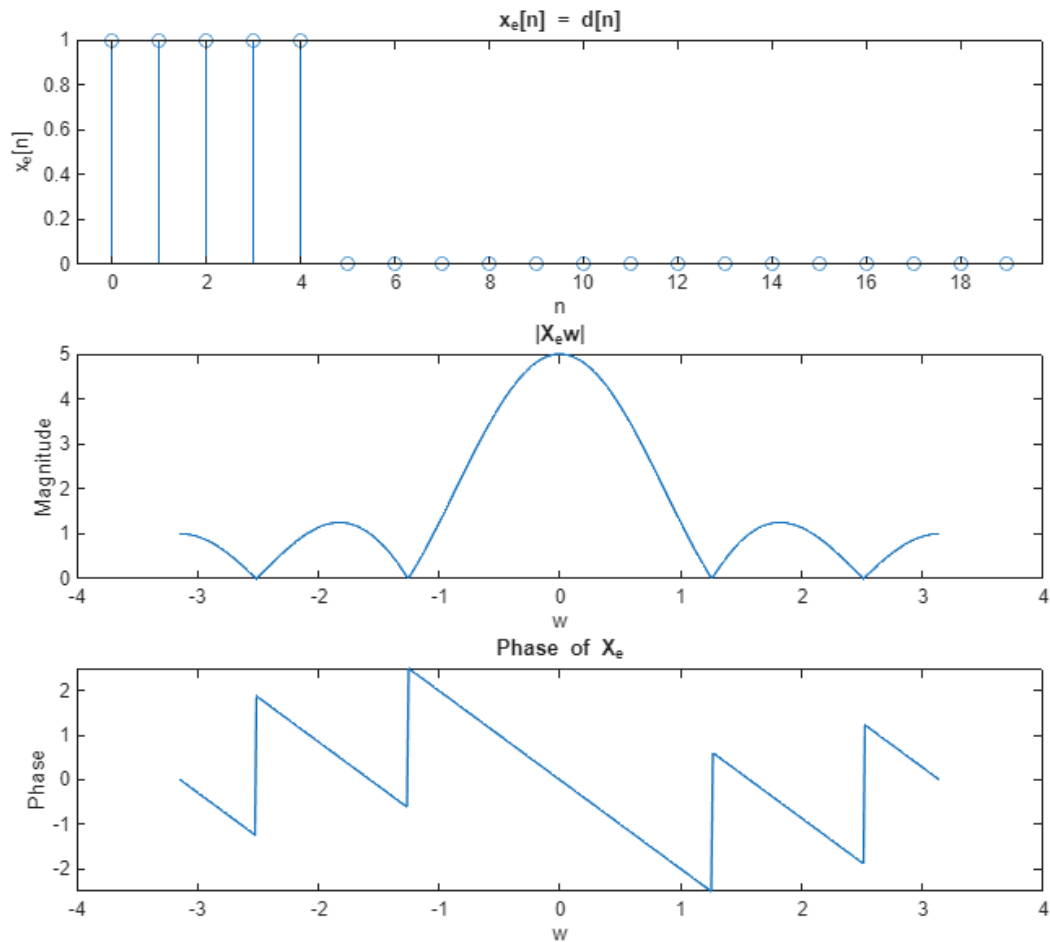
2 (e) PLOT DTFT

ALSO ANSWER: Is the data predominantly low frequency, high frequency, or neither?

```
x_e = [ones(1,5), zeros(1,15)];
H_e = DTFT(x_e, w);

figure;
subplot(3,1,1); stem(n, x_e); title('x_e[n] = d[n]'); xlabel('n');
ylabel('x_e[n]');
subplot(3,1,2); plot(w, abs(H_e)); title('|X_ew|'); xlabel('w');
ylabel('Magnitude');
subplot(3,1,3); plot(w, angle(H_e)); title('Phase of X_e'); xlabel('w');
ylabel('Phase');
disp('(e) This signal contains both low and high frequencies.');
```

(e) This signal contains both low and high frequencies.



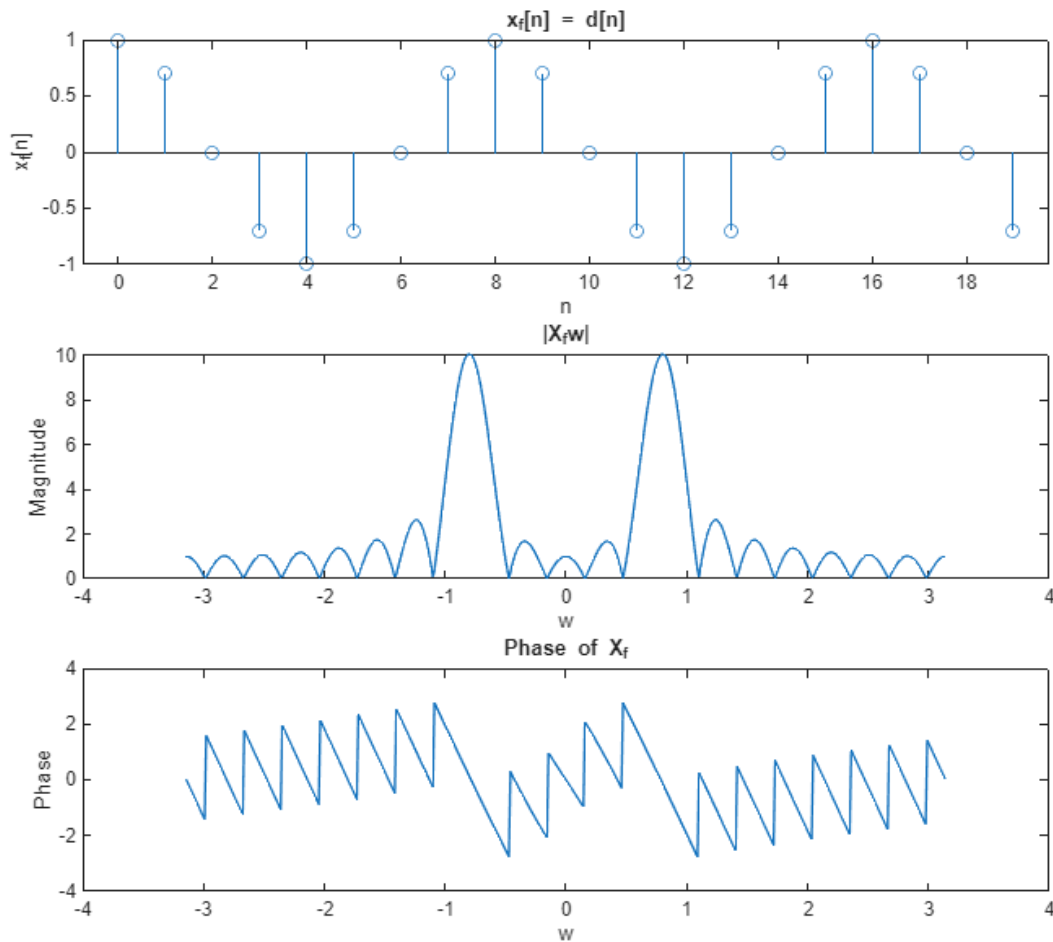
2 (f) PLOT DTFT

ALSO ANSWER: Is the data predominantly low frequency, high frequency, or neither?

```
x_f = cos(pi/4 * n);
H_f = DTFT(x_f, w);

figure;
subplot(3,1,1); stem(n, x_f); title('x_f[n] = d[n]'); xlabel('n');
ylabel('x_f[n]');
subplot(3,1,2); plot(w, abs(H_f)); title('|X_fw|'); xlabel('w');
ylabel('Magnitude');
subplot(3,1,3); plot(w, angle(H_f)); title('Phase of X_f'); xlabel('w');
ylabel('Phase');
disp('(f) This signal is predominantly low frequency.');
```

(f) This signal is predominantly low frequency.



QUESTION 3: DTFT PROPERTIES

3(a) PLOT DTFT

ALSO ANSWER: describe how each system changes the frequency domain

3(b) PLOT DTFT

ALSO ANSWER: describe how each system changes the frequency domain

3(c) PLOT DTFT

ALSO ANSWER: describe how each system changes the frequency domain

3(d) PLOT DTFT

ALSO ANSWER: describe how each system changes the frequency domain

3(e) PLOT DTFT

ALSO ANSWER: describe how each system changes the frequency domain

QUESTION 4: NULLING FILTER

```
% DO NOT REMOVE THE LINE BELOW
% MAKE SURE 'Noisy.wav' IS IN THE SAME DIRECTORY AS THIS FILE
[x, fs] = audioread('Noisy.wav');
```

4(a) EVALUATE DTFT OF INPUT SIGNAL

4(b) IDENTIFY FREQUENCY

```
% <== ANSWER TO QUESTION ==>
%
%
```

4(c) DESIGN FILTER

4(d) APPLY FILTER

```
% <== ANSWER TO QUESTION ==>
%
%
```

4(e) LISTEN TO AUDIO

ALL FUNCTIONS SUPPORTING THIS CODE

```
function H = DTFT(x,w)
% ===> Describe function here <===

    H = zeros(length(w),1);
    for nn = 1:length(x)
        H = H + x(nn).*exp(-1j*w.'*(nn-1));
    end

end
```

(a) $\delta[n]$ is neither high or low frequency.

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